Math is Boundless — Homework

Lesson 10-5 • Standard MPP.7

Name: _____ **Period:** _____ **Date:** _____

Today's Goals

Content: I can create and describe geometric designs using repetition, patterns, and rhythm, and use a pattern unit to predict what comes next.

Language: I can describe a design using the words repetition, pattern, pattern unit, and rhythm.

Key Words



repetition (repetición) — The same object, shape, or form repeated multiple times.



pattern (patrón) — A set of objects or shapes that are repeated and arranged in a regular order.



pattern unit (unidad del patrón) — The smallest chunk of a pattern that repeats.



rhythm (ritmo) — A design element created by varying the placement, size, orientation, or order of the same objects or shapes.



predictability (previsibilidad) — Being able to know what comes next because the design follows a rule.

Practice

Show your work. Use the blank space under each problem.

1. In geometric design, what is repetition?

A. A set of different shapes arranged randomly
B. Shapes that vary in size and order of appearance
C. Any drawing that uses color
D. The same object or shape repeated multiple times

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2. A border shows: pink circle, purple square, blue triangle, pink circle, purple square, blue triangle. What are the next three shapes?

A. Blue triangle, purple square, pink circle
B. Pink circle, purple square, blue triangle
C. Pink circle, blue triangle, purple square
D. It is impossible to predict

3. The pattern unit circle-square-triangle repeats. What is shape number 12?

A. The pink circle
B. The blue triangle
C. The purple square
D. It cannot be determined

4. Same pattern. What is shape number 20?

A. The purple square
B. The pink circle
C. The blue triangle
D. The 20th shape starts a new pattern

5. How many COMPLETE pattern units appear in the first 18 shapes of the circle-square-triangle border?

A. 3 units
B. 6 units
C. 9 units
D. 15 units

6. A painting shows circles whose colors, sizes, and thickness vary, along with the number of concentric circles inside each one. Describe how the painting demonstrates rhythm.

7. How is a design with rhythm different from a pattern?

A. Rhythm uses shapes and a pattern does not
B. A pattern varies its shapes and rhythm never does
C. There is no difference between them
D. Rhythm varies size, orientation, and order, so it often lacks predictability; a pattern repeats in a regular order

8. If a new row of circles were added to the rhythm painting, what would they look like? Explain your thinking.

9. The border pattern repeats circle, square, triangle. Name the shape at each position.

Position number	Thinking with the unit of 3	Shape
7	7 = 3 + 3 + 1, so it restarts	
12	12 = 3 × 4 exactly, ends a unit	
20	20 = 3 × 6 + 2, second in the	

Position number	Thinking with the unit of 3	Shape
	unit	